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Data and Work Sheets - Print or bring it on an electronic device

One-Dimensional Collisions - Physics 1401A
2026-2027

Please circle the appropriate values											
Course	**1101A**		**1201A**		**1401A**		**1501A**				
Lab Section	**002**	**003**	**004**	**005**	**006**	**007**	**008**	**009**	**010**	**013**	**014**
Lab Subsection	**A**	**B**	**C**	**D**							
Name	**First:** Beckett					**Last:** Graham					
Student #	2	5	1	5	5	0	6	5	6		
Lab Partner	**First:**					**Last:**					
Lab Station #											
Date	Nov 3 2026										
Demonstrator											

Disclaimer: Please note that some but not all questions in this lab writeup will be graded.

PART I: INVESTIGATIONS OF ELASTIC AND INELASTIC COLLISIONS

EXPERIMENT 1: PRELIMINARY ADJUSTMENTS OF THE TRACK, CARTS AND THE MOTION SENSORS

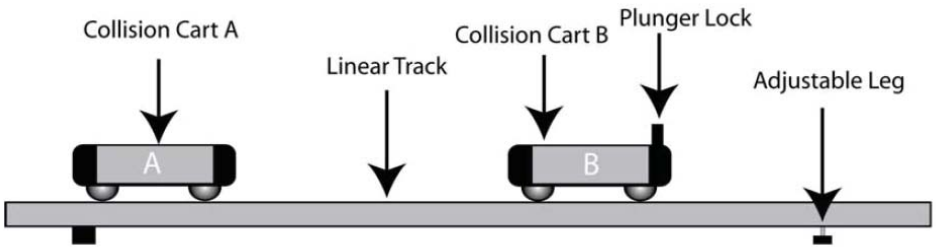


Figure 5: Linear track with collision carts.

Linear Track

The experiments in this lab utilize a smooth linear metal track.

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Levelling the Track

Center a bubble level on the track at the 60 cm mark. Use the adjustable levelling feet on the linear track (see Figure 5) to center the air bubble in the bubble level. Turn the bubble level 90 degrees to level the track in the perpendicular direction to travel. When the track is level a cart centered at the 60 cm mark should remain stationary.

Collision Carts

This lab uses two carts with low-friction ball bearing wheels that allow the carts to roll down the track with minimal resistance caused by friction. One end of each cart is labeled '**elastic**' and is used when performing elastic collision experiments. Magnets are mounted underneath the '**elastic**' ends of the carts in order to facilitate elastic collisions. The other end of the cart is not labeled but can be identified from the mounted velcro tabs, which are used to facilitate *inelastic* collisions.

Each cart is labeled either A or B. Cart B contains a spring-loaded plunger held in place by a 'plunger lock'. The plunger is used to facilitate the 'explosion' in Experiment 4. **Check that you have one of each type of cart as in Figure 5**.

Motion Sensors

Each experiment requires a measurement of the velocity of the carts before and after a collision. For this purpose you will use motion sensors mounted to the ends of the linear track as in Figure 7. The motion sensors emit pulses of ultrasound that reflect off ends of the carts and return to the motion sensors where they are detected and used to measure the position of the carts as a function of time. The *position vs time* information is sent to the computer where it is recorded and converted to *velocity vs time* data. The motion sensors have a switch on top with '*person*' or '*cart*' selections. Each motion sensor will produce smoother data in either one or the other mode, you should adjust your own sensors for best results.

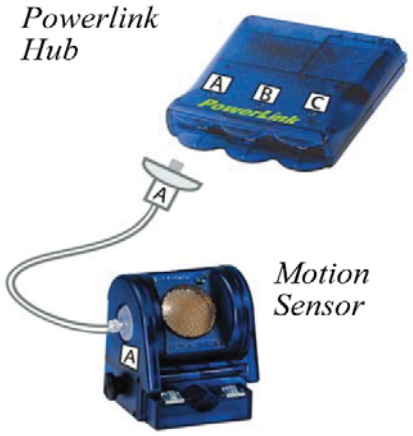


Figure 6: Connecting Motion Sensor A to port A on the *PowerLink* hub.

The motion sensors are labeled A, B or C, as shown in Figure 6. The ports on the *PowerLink* hub are also labeled A, B or C. Ensure that each motion sensor is connected to the correct port on the *PowerLink* hub.

The collision carts should be placed in each experiment such that they are closest to their matching motion sensor at the beginning of the experiment. In each experiment the motion sensors will begin collecting data when the velocity of one of the carts goes above v = 0.200 m/s. This helps to ensure good quality data with less noise in each experiment. If you see a message indicating that the initial conditions was not reached, try taking data again.

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Sign Convention

The velocity data produced by the motion sensors is given a sign based on whether the object under investigation is moving in the positive or negative x direction as shown in Figure 7. Carts moving to the right will be given a positive velocity while carts moving to the left will be given a negative velocity.

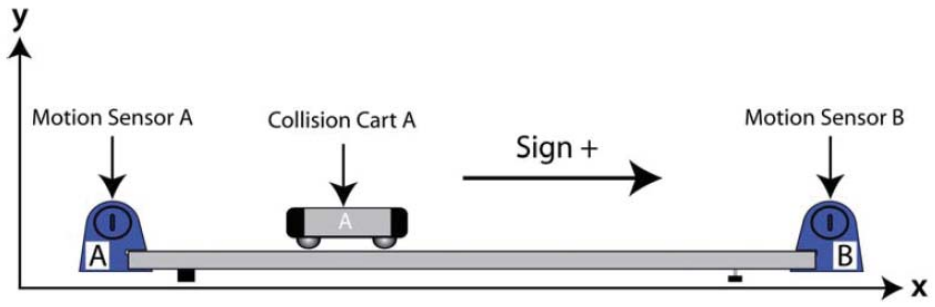


Figure 7: The experimental setup Experiment 1(a) showing the direction of motion that produces a positive velocity.

Note: *Ensure that your experimental setup is the same as shown in Figure 7, i.e. Motion Sensor A on the left side of the track, and Motion Sensor B on the right hand side of the track.*

EXPERIMENT 1(a): MOTION SENSOR ALIGNMENT

APPARATUS Linear track, two motion sensors, one collision cart, computer with *PASCO Capstone* software.

The alignment procedure is described as below.

Use the bubble leveler to make sure that the truck is perfectly horizontal. Make use of the adjustable legs if needed. Use the bubble leveler first along the track and subsequently perpendicular to the track.

Position your eye, level with the track, at the 50 cm mark. Your cheek should be almost touching the track with your eyes directed to motion sensor A. Adjust the tilt of the motion sensor so that you can see your reflection in the motion sensors reflective face. Follow the same procedure for motion sensor B.

This procedure is crucial to the rest of the experiment turning out well. Get help from your TA with this part if you need it!

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EXPERIMENT 1(b): HOW TO TAKE MEASUREMENTS

Open "1D Collisions.cap". Different pages will be available in this file. You can toggle among the pages by clicking on the heading names. The first page by default shows the experiment's outline. Click on the page video **"Instruction 1"** to learn how to use the *PASCO Capstone* graph interface and how to take measurements from a graph. Click the on the video to have it started. You can re-click on the video to pause the video as well. During the course of this video, the initial and final velocities of Cart A, from the data on the graph will be measured. Record the values given in the video below in Table 1.

Table 1: Velocities measured in the instructional video

Data Run	**Results**	
Run #1	Initial Velocity = 0.297 m/s	Final Velocity = 0.077 m/s

EXPERIMENT 1(c): ELASTIC COLLISIONS

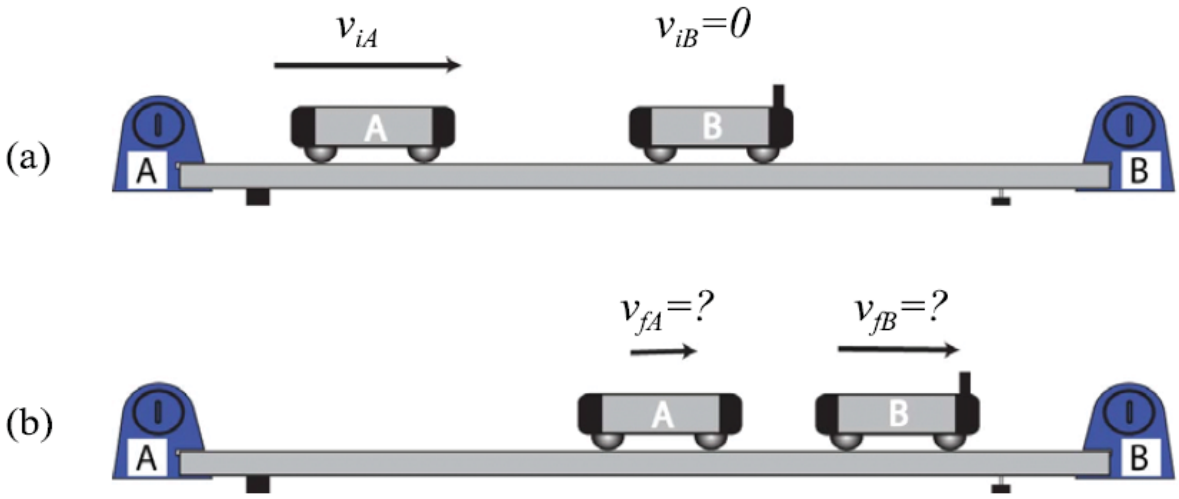


Figure 8: An elastic collision between two carts. Showing (a) before the collision: cart A has initial velocity $\mathbf{v}_{iA}$, and cart B is at rest, $\mathbf{v}_{iB} = 0$; (b) after the collision: cart A has final velocity $\mathbf{v}_{fA}$, and cart B has final velocity $\mathbf{v}_{fB}$

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OBSERVATION: ELASTIC COLLISION BETWEEN CARTS WITH EQUAL MASS

APPARATUS Linear track, two motion sensors, two collision carts, computer with *PASCO Capstone* software.

METHOD

(1) Watch the collision videos in the page heading "Collisions Video" - A Simple Experiment. Click ▶ to start a video.

(2) **Make a prediction** as to which video best represents an elastic collision between two carts of equal mass where one cart is initially at rest. Circle your prediction below.

Prediction:	(A)	(B)	(C)

Note: you will not lose marks if your prediction is wrong, but you will if you don't make a prediction!

(3) With the track satisfactorily levelled, place both collision carts on the track so that the sides of the carts labelled '**elastic**' are facing each other. With one cart stationary, give the other cart a gentle push so that it rolls towards the other at constant velocity. Watch how the velocities of each cart change after they collide. **Was your prediction correct?**

(Yes)	**(No)**

(4) Now, roll one cart slowly, and the other at least twice as fast, towards each other so that they collide. Observe again how the velocities of the carts change after they collide.

Question: Carts of equal mass collide in an elastic collision, where cart A has an initial momentum $\mathbf{p}_A$ and cart B has an initial momentum $\mathbf{p}_B$. What would you expect the final momenta of cart A and cart B to be?

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EXPERIMENT 2: INVESTIGATION OF AN ELASTIC COLLISION USING PASCO CAPSTONE SOFTWARE

APPARATUS Linear track, two motion sensors, two collision carts, metal bar, electric balance, computer with *PASCO Capstone* software.

INTRODUCTION

To study the law of conservation of momentum for an elastic collision, you will perform a series of experiments to investigate how the initial velocity and mass of each cart involved in a collision affects the final velocities of the carts after a collision, thus we will study

METHOD

Click on the **"Elastic Collision Page"**. Measure the mass of each cart and the metal bar using the electronic balance and record your results in the section Data before Table 2. The mass of the carts are very similar and you will place the metal bar on cart A to increase its mass. In this experiment, *PASCO Capstone* will provide graphs of the *total momentum vs time* and *total kinetic energy vs time*. In order for *PASCO Capstone* to provide these graphs, you must register the mass of each cart in *PASCO Capstone*.

To register the masses of the carts in *PASCO Capstone*, click the Calculator button  on the *PASCO Capstone* left side of the screen under Tools. A window will pop up. Select and double-click M_1 from the list. Enter your measured value for the total mass of cart A (cart + metal bar) in kg. This will be the mass of cart A in the experiment. Select and double-click M_2. Enter the mass of cart B and then click the accept button  . **Be sure not to click the delete button by mistake**.

Center Cart B at around the 70 cm mark on the linear track with it's 'elastic' end facing motion sensor A. With cart B stationary, hit the *PASCO Capstone* record button  and give cart A a push towards cart B to begin collecting data (as in Figure 8). The computer screen will display the *velocity vs time*, *total momentum vs time* and *total kinetic energy vs time* graphs for both cars. Measure the initial velocity $\mathbf{v}_{iA}$ of cart A, the final velocities $\mathbf{v}_{fA}$ and $\mathbf{v}_{fB}$ of carts A and B respectively. Also measure the total initial momentum $\mathbf{p}_i$, total final momentum $\mathbf{p}_f$, total initial kinetic energy K_i and total final kinetic energy K_f for carts A and B. Record the measured values in Table 2. Repeat the experiment two more times. To delete previously taken plots, go to 'Experiment' menu. Note, in *PASCO Capstone*, if the entire plot area (3 plots) is not fully visible within the screen area, click on the graph and press the scale axes button  to rescale your screen resolution.

DATA

Mass of cart A 271.1g

Mass of the metal bar _____

Total mass of cart A = mass of cart A + metal bar _____

Mass of cart B _____

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Table 2: Experiment 2: Elastic collision measurements for two carts with different masses

Trial	$\mathbf{v}_{iA}$ (m/s)	$\mathbf{v}_{iB}$ (m/s)	$\mathbf{p}_i$ (kg m/s)	K_i (J)	$\mathbf{v}_{fA}$ (m/s)	$\mathbf{v}_{fB}$ (m/s)	$\mathbf{p}_f$ (kg m/s)	K_f (J)	% Diff p_i & p_f	% Diff K_i & K_f
1										
2										
3										

Do not delete your last data run for this activity. You will need these graphs to answer a post-lab question at the end of this manual.

The last two columns in Table 2 show the percentage difference between the initial and final values of momenta and kinetic energy.

CALCULATIONS

Sample Calculation: linear momentum before and after the elastic collision
Select one of the trials from Table 2 and provide a sample calculation for the initial and final momenta values ($\mathbf{p}_{iA}$, $\mathbf{p}_{fA}$ and $\mathbf{p}_{fB}$) of the collision carts. Record the trial number. Use SI units.

Trial #: _____

$\mathbf{p}_{iA}$ _____

$\mathbf{p}_{fA}$ _____

$\mathbf{p}_{fB}$ _____

Total initial momentum of the two carts $\mathbf{p}_i$ _____

Total final momentum of the two carts $\mathbf{p}_f$ _____

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For the trial you have selected, compare the initial and final total momentum of the carts using the percent difference test.

$$\textbf{\% difference test} = \frac{\left|\mathbf{p}_f - \mathbf{p}_i\right|}{\left(\mathbf{p}_i + \mathbf{p}_f\right)/2} \times 100 < 10\%$$

What is your conclusion about the total linear momentum of the two carts?

Sample Calculation: kinetic energy before and after the elastic collision

Provide a sample calculation for the initial kinetic energy K_{iA} and the final kinetic energies K_{fA} and K_{fB} of the collision carts. Use the data from the same trial used in the previous sample calculation.

K_{iA} _____

K_{fA} _____

K_{fB} _____

Total initial kinetic energy of the two carts K_i _____

Total final kinetic energy of the of the two carts, K_f _____

For the trial you have selected, compare the initial and final total kinetic energies of the carts.

$$\% \text{ difference between } K_i \text{ and } K_f = \frac{\left|K_f - K_i\right|}{\left(K_i + K_f\right)/2} \times 100 < 10\%$$

What can you conclude about the total kinetic energy of the two carts?

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EXPERIMENT 3: INVESTIGATION OF AN INELASTIC COLLISION

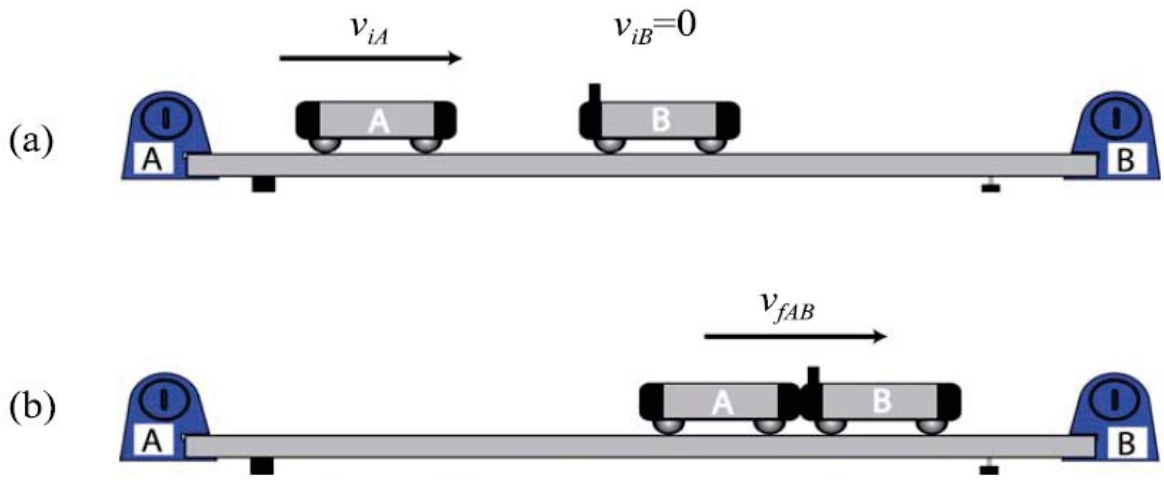


Figure 9: An inelastic collision between carts A and B. (a) Before the collision, cart B is at rest. (b) After the collision the two carts stick together, travelling with a velocity $\mathbf{v}_{fAB}$.

APPARATUS Linear track, two motion sensors, two collision carts with velcro pads, computer with *PASCO Capstone* software, electric balance.

INTRODUCTION

In this experiment, you will study the linear momentum and kinetic energy of two carts undergoing complete inelastic collisions. In an inelastic collision, the total linear momentum is conserved while the total kinetic energy is not conserved.

Uncertainties

Estimate the *absolute uncertainties* in your measurements from the standard deviation (std. dev.) in your measurements. In each measurement, your standard deviation should have a magnitude no greater than 0.003. Get help from your TA if it is more than this.

METHOD

Click on the ”Inelastic Collisions” Page.

Using the Calculator button on the *PASCO Capstone* menu, change the mass of Cart A back to the mass of the cart without the mass bar on top. After entering the mass, be sure to click the ‘accept’ button. Place cart B at rest, centered at the 70 cm mark on the track. Arrange cart B so that it’s ‘inelastic’ end is facing motion sensor A. Place cart A near the motion sensor A with it’s ‘inelastic’ end facing cart B as in Figure 9(a). Press the *PASCO Capstone* record button and then push cart A towards cart B so that the carts stick together when they collide as in Figure 9(b). From *PASCO Capstone* plots, measure the velocity of the carts just before and just after the collision. Record your results in Table 3.

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PASCO Capstone will use the velocity data and the mass measurements entered as experimental constants to produce graphs of *total momentum vs time* and *total kinetic energy vs time*. From these plots displayed on the computer, measure the total momentum just before the collision $\mathbf{p}_i$ and just after the collision $\mathbf{p}_f$, as well as the total kinetic energy just before the collision K_i and just after the collision K_f. Take three data runs, and record your results in Table 3.

DATA

Table 3: Inelastic collision measurements for two carts.

Mass of cart A, m_A _____, Mass of cart B, m_B _____

Trial	$\mathbf{v}_{iA}$ (m/s)	$\mathbf{v}_{iB}$ (m/s)	$\mathbf{p}_i$ (kg m/s)	K_i (J)	$\mathbf{v}_{fAB}$ (m/s)	$\mathbf{p}_f$ (kg m/s)	K_f (J)	% Diff p_i & p_f	% Diff K_i & K_f	$(K_i\text{-}K_f)/K_i$
1										
2										
3										

CALCULATIONS

Linear momentum before and after an inelastic collision:
What can you conclude about the total linear momentum of the two carts?

Kinetic energy before and after an inelastic collision:
What can you conclude about the total kinetic energy of the two carts?

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Fractional kinetic energy lost $(K_i - K_f)/K_i$

The last column of Table 3 gives the fraction of kinetic energy lost compared to the initial kinetic energy. What happens to the lost energy in this situation?

Table 3 shows three inelastic collisions with varying velocities and kinetic energies but for the same cars with constant mass. What can you say about the fraction $(K_i - K_f)/K_i$ for the three collisions?

PART II: INVESTIGATIONS OF EXPLOSIONS AND IMPULSE IN ONE-DIMENSIONAL COLLISIONS

In Experiment 4, you will study the conservation of momentum in an explosion when there are no external forces acting on the system. In Experiment 5, you will take data to verify that the impulse J or the change in momentum during a collision is given by $J = p_f - p_i$.

EXPERIMENT 4: CONSERVATION OF LINEAR MOMENTUM IN ONE-DIMENSIONAL EXPLOSIONS

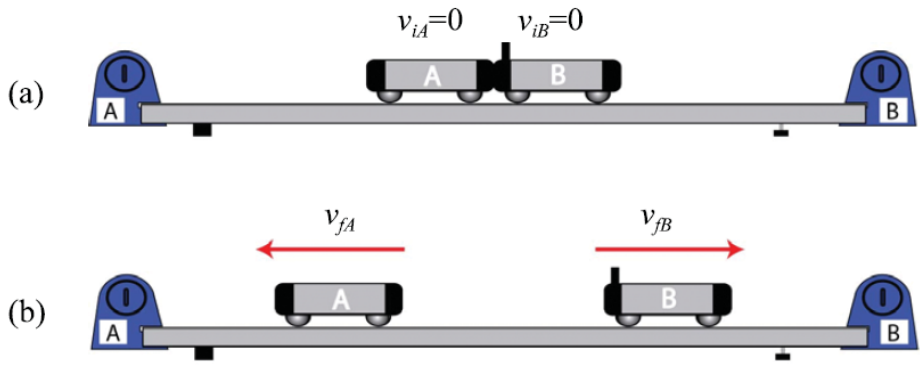


Figure 10: One-Dimensional Explosion. (a) Before explosion, carts A and B are together at rest. (b) After explosion, carts A and B move in opposite directions.

APPARATUS Linear track, two motion sensors, two collision carts, metal bar, computer running *PASCO Capstone*.

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INTRODUCTION

In this experiment, you will study the momentum of each cart involved in an explosion.

Uncertainties

Estimate the *absolute uncertainties* in your measurements from the standard deviation (std. dev.) in your measurements. In each measurement, your standard deviation should have a magnitude no greater than 0.003 in the units of the quantity of interest. Get help from your TA if it is more than this.

METHOD

Click on the "1D Explosion" Page. One of the collision carts that you are using will have a small tab sticking out the top of it. Pressing the tab down will release a spring-loaded plunger from the end of the cart. The plunger will be used in this experiment as the internal impulse for the explosion. Load the plunger by pushing it in and slightly upwards until it catches. Don't force it! It should load easily and smoothly. Place carts A and B together in the middle of the linear track such that the '*inelastic*' ends of the carts are touching. Hit the *PASCO Capstone* record button  , to begin collecting data. Set off the explosion by tapping the tab on top of the cart with the spring loaded plunger (tapping the tab lightly with the metal bar works well).

Measure the final velocities, $\mathbf{v}_{fA}$ and $\mathbf{v}_{fB}$, of carts A and B, respectively, using the *PASCO Capstone* statistics tool as previously described. Record the standard deviation as the uncertainty in the measurement. Record your data in Table 4. Perform the experiment a total of three times.

DATA AND CALCULATIONS

Table 4: Data for one-dimensional explosion.

Mass of cart A _____, Mass of cart B _____

Trial	$\mathbf{v}_{fA}$ (m/s)	$\mathbf{v}_{fB}$ (m/s)	$\mathbf{p}_{fA}$ (N s)	$\mathbf{p}_{fB}$ (N s)	$\mathbf{p}_f$ (N s)	% Difference $\lvert\mathbf{p}_{fA}\rvert$ & $\lvert\mathbf{p}_{fB}\rvert$
1						
2						

What can you conclude from the results in Table 4?

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EXPERIMENT 5: CHANGE OF MOMENTUM AND IMPULSE IN ELASTIC COLLISIONS AND IMPULSE

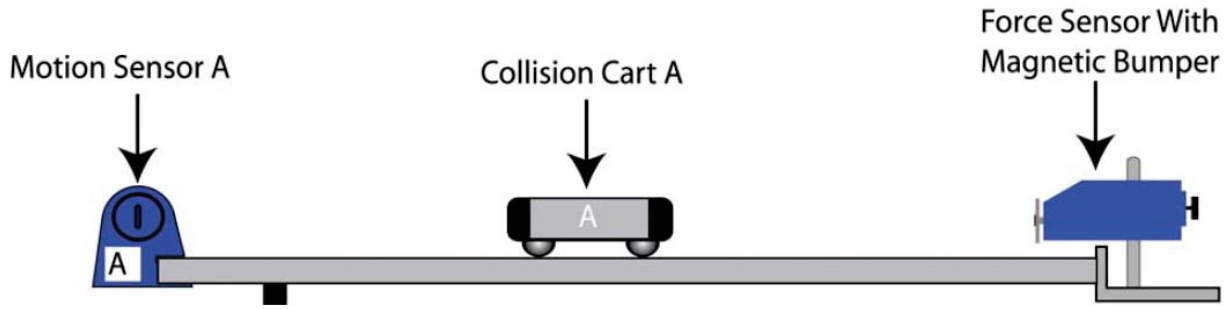


Figure 11: Setup for measuring the impulse and change of momentum in Experiment 5.

APPARATUS Linear track, motion sensor, collision cart, force sensor with magnetic bumper, computer running *PASCO Capstone*, pan balance.

INTRODUCTION

In Experiment 5 the change of momentum that an cart undergoes during an elastic collision is measured. The forces that occur during the collision are measured as a function of time and related to the change in momentum experienced by the collision cart.

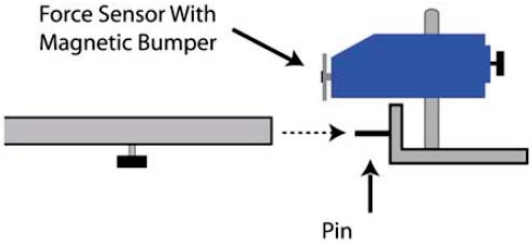


Figure 12: Inserting the linear track onto the force sensor bracket.

METHOD

The video "Instruction 2" explains how to measure the area under the force vs time curve. Use cart A for this experiment. Remove motion sensor B from the linear track. Mount the linear track onto the bracket holding the force sensor and magnetic bumper on the right side of the table. The two pins on the bracket slide into slots on the end of the linear track to secure it in place as shown in Figure 12 (note: do not undo the clamp holding the force sensor and bracket to the table). If setup properly, the apparatus should look like that in Figure 11.

Click on "Change of Momentum and Impulse" Page.

Important: *Press the 'zero' button on the top of the force sensor before each data run.*

Important: *PASCO Capstone will use the mass of cart A in it's calculations. Ensure that you use collision cart A for this experiment.*

Position the collision cart so it is centered on the 20 cm mark on the track with the elastic end of the cart facing the force sensor. Click the *PASCO Capstone* record button,  , to begin collecting data then push the cart so that it rolls towards the force sensor and magnetic bumper. Push the *PASCO Capstone* record button again to stop the recording, after the cart bounces back from the magnetic bumper.

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You should now have a smooth graph of *velocity vs time*, *momentum vs time* and *force vs time*. If the cart was pushed too hard, the graph of *force vs time* will have a sharp spike at the centre. In this case, erase the last data run and repeat.

Determine the initial velocity $\mathbf{v}_i$ and momentum $\mathbf{p}_i$ and the final velocity $\mathbf{v}_f$ and momentum $\mathbf{p}_f$ from the *PASCO Capstone* graphs. Enter the results in Table 5.
The area under the *force vs time* graph is equal to the total impulse *J*. *PASCO Capstone* measures the area under the entire *force vs time* curve and lists it in the graph legend. The default selection for the area under the curve will include the whole curve which includes 'noise' that the force sensor picks up when there is no force applied to the sensor. To get a more accurate value for the area under the *force vs time* curve, click and drag a square over the area that corresponds to the duration over which the collision occurs. Enter the results in Table 5.

DATA AND CALCULATIONS

Table 5: Velocities of the Collision Cart and Impulse.

Mass of cart A, m_A _____

Trial	Initial Velocity $\mathbf{v}_i$ (m/s)	Final Velocity $\mathbf{v}_f$ (m/s)	Initial Momentum $\mathbf{p}_i$ (kg m/s)	Final Momentum $\mathbf{p}_f$ (kg m/s)	Change in Momentum Δp (kg m/s)	Impulse J (kg m/s) (Area under F *vs* t graph)	% Difference between $\lvert\Delta p\rvert$ and $\lvert J\rvert$
1							
2							

What conclusion can you draw from the results in the last column?

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Post-Lab Questions

(1) The human body is injured not by high speeds but by high forces associated with high accelerations. Using the data already collected in Experiment 2, compute the acceleration felt by both carts during the elastic collision. Compare the accelerations of the more massive cart to the less massive cart. Which cart had the higher acceleration? Extrapolating up in size, would you expect a passenger to be safer in a car collision in a heavier car or a lighter one?

(2) The momentum of a car crashing into a wall will change very quickly as the car decelerates rapidly to a speed of zero. Using the equation for change in momentum, $\Delta p = F \cdot \mathrm{d}t$, explain how can an air bag effectively protect a passenger during a car crash?

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CONCLUSIONS AND DISCUSSION

(In this section, summarize the conclusions from Experiments 2 to 5. What physics concepts did you study? Did you get the expected results? If not, why? What are the sources of errors in this lab?)